

Name: _____





BOOTSTRAP

Equity • Scale • Rigor

Brought to you by the Bootstrap team:

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Exploring the Driving Simulator

Open [Self-Driving Car Starter File](#) in Pyret and save a copy of your own. Then click "Run".

Playing the Game

1) **Test run:** evaluate `train-bev(20)` to run the simulator at a speedy 20 frames per second. Use the left- and right-arrow keys to steer the car. In the top-left corner of the simulator, you'll find the current `steering-angle`, along with some other data.

2) What happens to the track when you cross the finish line? _____

3) When you close the training window, or press "escape", you should see a table. What do you Notice? What do you Wonder? _____

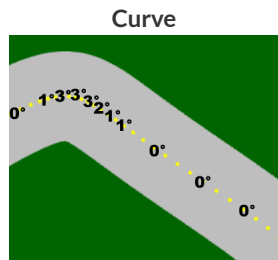
Every time we drive the car, we create a new table based on how we steer. But to *save that data* and use it later, we need to define it.

4) **Training run:** slow down the training lap to get better data, and **save it as a variable**. Evaluate `run1 = train-bev(10)` and steer the car to the finish the line. When you close the simulator, evaluate `run1`. How many rows are in your table? _____

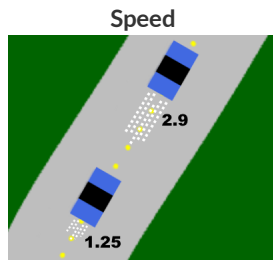
5) Now let's try another lap using a "Point of View" simulation. Evaluate `run2 = train-pov(5)`. Which view do you prefer, and why? _____

Examining the Training Data

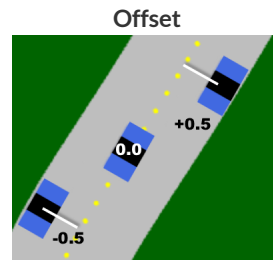
Each row in your table represents one snapshot of your drive, showing the car's state at that instant.



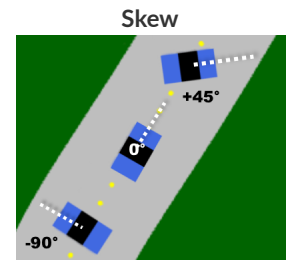
The curvature of the road in front of the car (-6° to 6°)



How fast the car is moving (2 to 3)



How far the car is from the center line (-45 to 45)



Angle of the car relative to the road (-135° to 135°)

6) Which column do you think is most important for predicting `steering-angle`? Why? _____

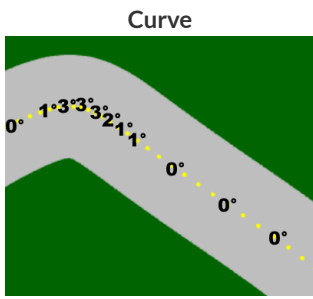
The Challenge of Training

7) What might happen if a self-driving car traveled on a *curved* road, after being trained on data collected on a perfectly *straight* road? _____

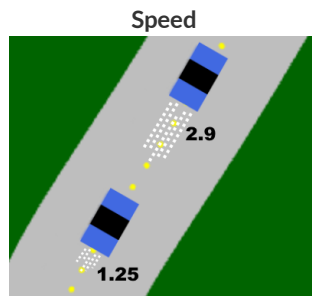
8) Describe your **ideal** training dataset: how many rows does it have? Is it trained on different tracks? Speeds? What else? _____

What Predicts Steering Angle?

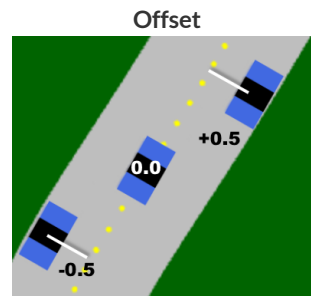
There are four explanatory variables in our dataset:



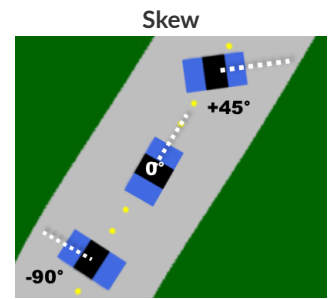
The curvature of the road in front of the car



How fast the car is moving



How far the car is from the center line

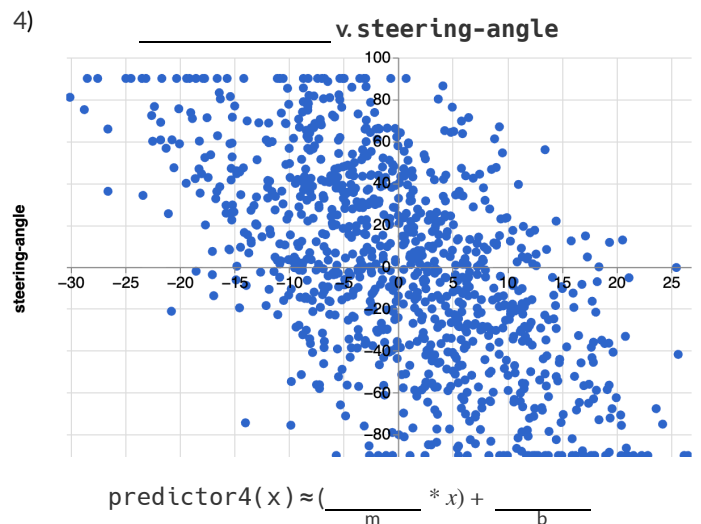
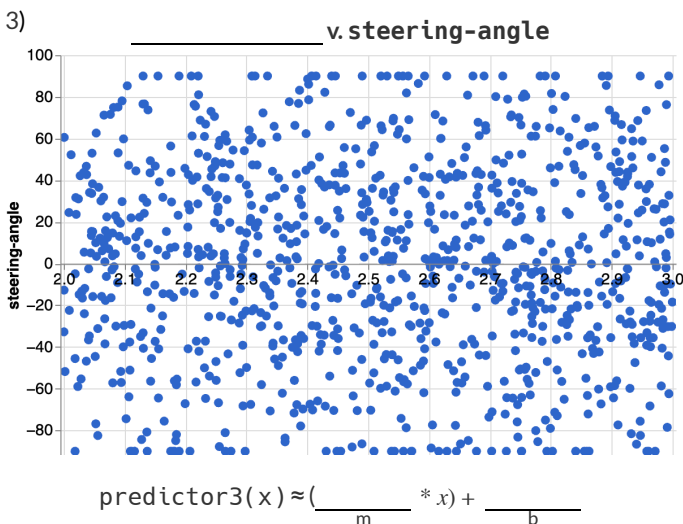
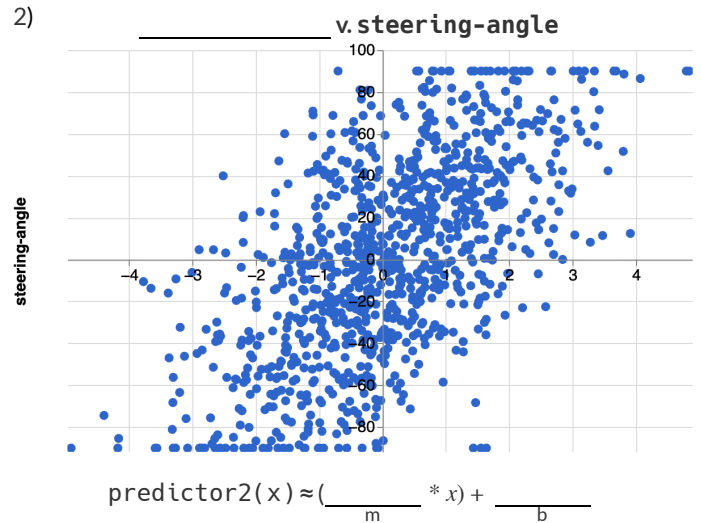
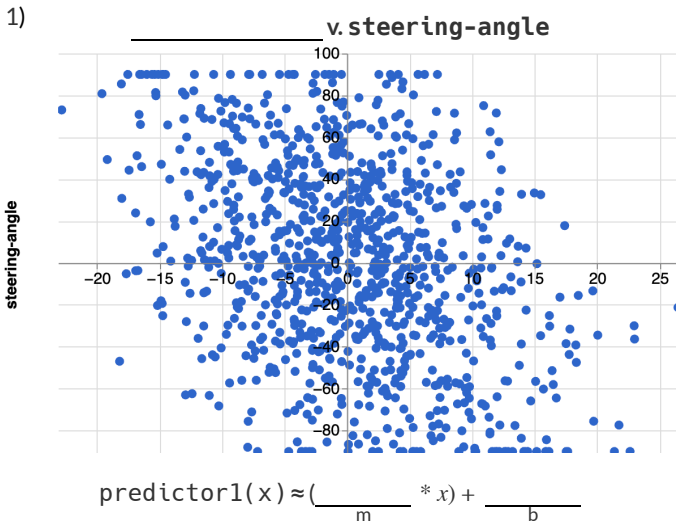


Angle of the car relative to the road

Make a scatter plot of steering-angle v. each of these variables in [Self-Driving Car Starter File](#).

For each of the plots below:

- **Match** them to the scatter plots you made in Pyret to fill in the titles with the missing variables
- **Draw** what you think the model (line of best fit) would look like and **estimate** its slope and y-intercept



5) What two models do you think will be the best predictors of steering-angle? _____

Building a Predictor for Steering Angle

This page is designed for use with the [Self-Driving Car Starter File](#)

Using Linear Regression

1) `lr-plot(training, "id", "curve", "steering-angle")` was used to fill in the first row below. Complete the table.

Explanatory Variable	Slope	Intercept	S-value
curve	19.5	-0.4	37.0
offset			
skew			
speed			

2) How do Pyret's best-fit lines compare to your estimates on [What Predicts Steering Angle?](#) _____

3) Use the values you calculated with `lr-plot` for `curve` to define `c-predictor` in math and Pyret notation below. Then complete the template in the starter file and click "Run".

$c\text{-predictor}(curve) = (\text{ } \times curve) + \text{ }$

`fun c-predictor(curve): (   * curve) +   end`

4) On a straight road (`curve = 0`), `c-predictor` produces a steering angle of (round to the nearest degree) $\frac{\text{ } \text{degrees}}{\text{intercept} \text{ units}}$. Does this make real-world sense? _____

5) If the `curve` of the road increases by 1°, the `c-predictor` expects the steering angle to $\frac{\text{ } \text{increase/decrease}}{\text{slope}}$ by $\frac{\text{ } \text{slope}}$.

6) To generalize what the model *learned*: "the more sharply the road curves, the $\frac{\text{ } \text{less? more?}}{\text{ }}$ sharply you need to steer".

Driving with the Model

```
# drive :: (predictor :: Row -> Number) -> Table
# runs the driving simulator, using predictor to compute the steering angle at each frame
```

7) Read the Domain of `drive` carefully. What do you Notice? _____

8) Evaluate `drive(c-predictor)` and read the error carefully. What do you think it is telling us? _____

Evaluate `regression-model-code(training, [list: "curve"], "steering-angle")` and read the resulting code carefully.

9) How is the resulting code **similar** to your `c-predictor` function? _____

10) How is it **different**? _____

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